

Employee Dashboard - SDD & PDD

Employee Dashboard — Power BI

Combined System Design Document (SDD) & Project Design Document (PDD)

Prepared by: Madhumitha **Platform:** Microsoft Power BI Desktop

System Design Document (SDD)

Employee Dashboard — Power BI

Project	Employee Dashboard
Prepared by	Madhumitha
Platform	Microsoft Power BI Desktop
Document type	System Design Document

1. Introduction

1.1 Purpose

This document describes the technical design of the Employee Dashboard — its architecture, data model and report layout — so the system can be understood, maintained and extended from the GitHub repository.

1.2 Scope

Covers the Power BI .pbix file only: the cleaned data table, the visuals built on it, and the report page. It does not cover the original raw HR source file before cleaning.

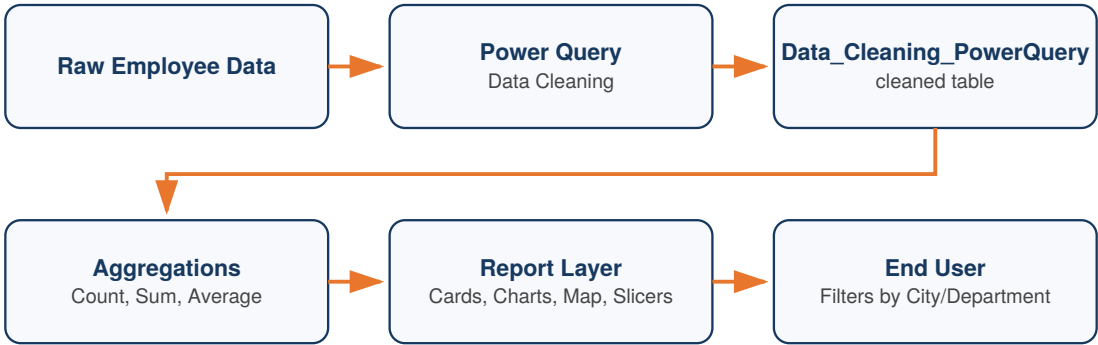
1.3 Definitions

Term	Meaning
SDD	System Design Document (this document)
PDD	Project Design Document (companion document, see PDD.md)
Power Query	Power BI’s data cleaning/transformation engine

2. System Overview

The dashboard gives HR stakeholders a single-page, filterable view of the workforce — headcount, total and average salary, department mix, performance rating by department, and employee location — filterable by City and Department.

3. System Architecture



System Architecture

Layers 1. **Raw source data** — original employee records (assumed to arrive with inconsistencies). 2. **Cleaning layer** — Power Query transforms the raw data into a single analysis-ready table, named `Data_Cleaning_PowerQuery` to make the cleaning step traceable. 3. **Aggregation layer** — visual-level aggregations (count, sum, average) computed directly on the cleaned table. 4. **Report layer** — one report page presenting the aggregations visually. 5. **Consumption** — the end user filters via slicers; the underlying table is not edited.

4. Data Model

4.1 Table

Table	Role
Data_Cleaning_PowerQuery	Single cleaned fact table holding one row per employee

4.2 Data Dictionary — columns used by the report

Column	Type (inferred)	Used for
EmpID	Identifier	Headcount (Count of EmpID)
Salary	Numeric	Total salary (card), average salary (combo chart)
Department	Text	Donut chart, bar chart, combo chart, slicer
City	Text	Map, slicer
PerformanceRating	Numeric	Average performance rating by department (bar chart)

This is a single flat table rather than a star schema — appropriate given the dashboard’s scope (one entity: the employee).

5. Measures / Aggregations

Aggregation	Definition	Visual
Count of EmpID	Distinct/non-null count of EmpID	KPI card (headcount)
Sum of Salary	Sum of Salary	KPI card (total salary)
Average of Salary	Average of Salary by Department	Combo chart
Average of PerformanceRating	Average of PerformanceRating by Department	Bar chart

These are implicit (visual-level) aggregations rather than named DAX measures — each visual defines its own aggregation directly on the column.

6. Report / UI Design

Visual	Type	Fields	Purpose
Header	Text box	“EMPLOYEE DASHBOARD”	Page title
KPI cards	Card visual	Count of EmpID, Sum of Salary, Department	Headline workforce totals
Department split	Donut chart	Department, Count of EmpID	Headcount share by department
City slicer	Slicer	City	Filter by location
Performance by department	Clustered bar chart	Department, Average of PerformanceRating	Compare performance across departments
Department slicer	Slicer	Department	Filter by department
Salary trend	Line & stacked column combo chart	Department, Average of Salary	Compare average salary across departments
Location map	Map	City	Geographic distribution of employees

All visuals read from the same table, so the City and Department slicers filter every visual on the page together.

7. Non-Functional Requirements

- **Usability** — single page, two slicers, no nested navigation.
 - **Performance** — one flat, pre-cleaned table keeps aggregations fast even as row counts grow.
 - **Maintainability** — the `Data_Cleaning_PowerQuery` naming makes it explicit where cleaning happens, so future edits go to Power Query rather than the report.
 - **Portability** — self-contained `.pbix` file; no external services required to open it.
-

8. Tools & Technology Stack

- **Power BI Desktop** — data cleaning (Power Query), modeling and report design
 - **Power Query (M)** — data cleaning/transformation prior to load
-

9. Assumptions & Constraints

- Raw HR data is assumed to have required cleaning (missing values, formatting) before this table was produced; the cleaning logic itself lives in Power Query and isn't visible from the report layer.
- The report is a single page; it is not built for drill-through or multi-page navigation.

Project Design Document (PDD)

Employee Dashboard — Power BI

Project	Employee Dashboard
Prepared by	Madhumitha
Document type	Project Design Document

1. Project Overview

1.1 Problem Statement

Raw employee data (headcount, salary, department, location, performance) is hard to read as records — HR stakeholders need a visual, filterable view to understand workforce composition and spot patterns quickly.

1.2 Objective

Deliver a single Power BI dashboard that shows headcount, salary, department distribution, performance rating and employee location in one place, filterable by City and Department.

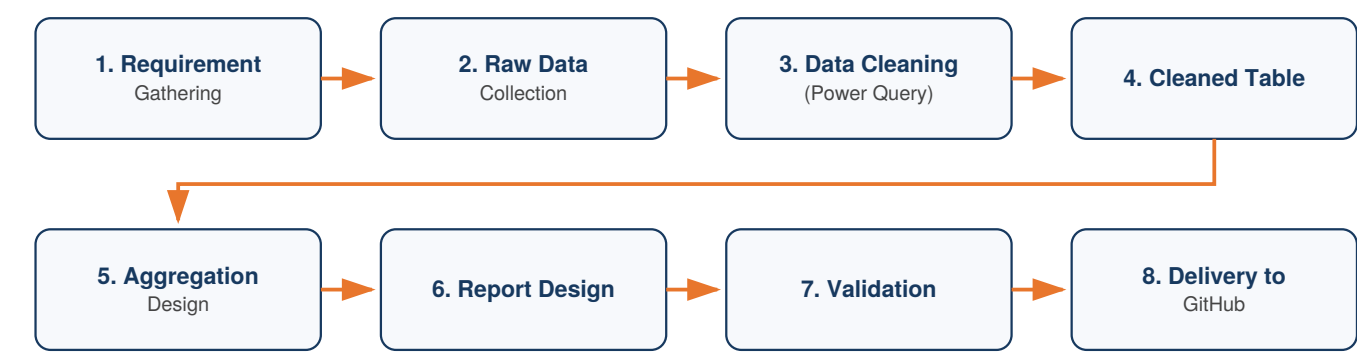
1.3 Scope

In scope: data cleaning, headcount and salary KPIs, department-wise distribution and performance, city-wise employee mapping, filtering by City and Department. **Out of scope:** payroll processing, recruitment/ATS data, predictive attrition modeling, multi-page drill-through, scheduled automated refresh.

2. Stakeholders

Stakeholder	Interest
HR Leadership	Overall workforce composition and cost (headcount, salary)
Department Heads	Department-wise headcount, salary and performance
HR Analysts	Explore data by city/department for reporting
Project Owner (Madhumitha)	Build, validate and maintain the dashboard

3. Project Methodology / Process Flow



Project Process Flow

Step	Description
1. Requirement gathering	Define which workforce questions HR needs answered
2. Raw data collection	Gather employee records (ID, salary, department, city, performance)

3. Data cleaning	Clean and shape the raw data in Power Query
4. Cleaned table	Load into a single analysis-ready table
5. Aggregation design	Decide which counts/sums/averages each visual needs
6. Report design	Lay out KPI cards, donut, bar chart, combo chart, map and slicers
7. Validation	Check visual totals against the raw source figures
8. Delivery	Publish the final .pbix and supporting docs to GitHub

4. Deliverables

Deliverable	Description
EMPLOYEE_POWERBI_DASHBOARD.pbix	The Power BI report file
SDD.md	System Design Document (architecture, data model, visuals)
PDD.md	This Project Design Document (process, scope, roles)

5. Roles & Responsibilities

Role	Responsibility	Owner
Data Analyst / Developer	Data cleaning, aggregation design, report build	Madhumitha
Reviewer	Validate figures against source data	Madhumitha

6. Risks & Mitigation

Risk	Mitigation
Raw data has missing/inconsistent values	Handled explicitly in the Power Query cleaning step
Single flat table limits future scalability (e.g. adding a time dimension)	Documented as a known constraint; a star schema can be introduced later if needed
Department/City names inconsistent (typos, casing)	Standardize during cleaning before load

7. Success Criteria

- Headcount, total salary and department mix are visible at a glance without scrolling.
- City and Department slicers correctly filter every visual on the page.
- KPI and chart figures reconcile with the raw employee data.

8. Future Enhancements

- Add a time dimension for headcount/salary trend over time.
- Attrition and tenure analysis.
- Drill-through page for a single department or city.